

Cancer drug treatment for myeloma

Myeloma is usually treated with a combination of different cancer drugs. This often includes targeted therapy drugs and a steroid. You may also have these drugs with chemotherapy.

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Cancer drugs for myeloma

[Myeloma](#) is usually treated with a combination of different cancer drugs. This often includes targeted therapy drugs and a steroid. You may also have these drugs with chemotherapy.

Drug combinations are often named after the initials of the drugs used. The following are some combinations used to treat myeloma:

- [PVD](#) – panobinostat, bortezomib (Velcade[®]) and dexamethasone
- [IRD](#) – ixazomib, lenalidomide (Revlimid[®]) and dexamethasone
- [DVTD](#) – daratumumab, bortezomib (Velcade[®]), thalidomide and dexamethasone
- [DVCD](#) – daratumumab, bortezomib (Velcade[®]), cyclophosphamide and dexamethasone
- [DRD](#) – daratumumab, lenalidomide (Revlimid[®]) and dexamethasone
- [KRD](#) – carfilzomib (Kyprolis[®]), lenalidomide (Revlimid[®]) and dexamethasone
- [SVD](#) – selinexor, bortezomib (Velcade[®]) and dexamethasone.

Your doctor and nurse will talk to you about the different treatment options. They will tell you:

- the aims of the treatment
- [how it is given](#)
- how often you have it (treatment cycle)
- how long your course will take
- any possible side effects.

During a course of treatment, you have regular [blood tests](#) to check the effect of the drugs. The dose of the drugs may be changed based on the results of your blood tests, or any side effects.

Cancer drugs can cause side effects. It is important to tell your cancer doctor or specialist nurse if you have any side effects while you are having treatment. They may change your treatment to control side effects or to help to prevent more serious problems.

Targeted therapy for myeloma

Targeted therapies are drugs that affect the way myeloma cells grow. Some targeted therapy drugs affect the immune system and help it to destroy myeloma cells. These are called immunotherapy drugs.

The targeted therapies used for newly diagnosed myeloma are:

- [thalidomide](#), [lenalidomide](#) or [pomalidomide](#)
- [bortezomib](#)
- [daratumumab](#)
- panobinostat
- ixazomib
- [carfilzomib](#)
- selinexor.

You usually have these drugs with chemotherapy drugs and [steroids](#). Sometimes you have 2 targeted therapy drugs together.

If you are taking thalidomide, lenalidomide or pomalidomide, you will be given drugs called anti-coagulants to thin your blood. This is because these drugs can increase your risk of developing a [blood clot](#). Anti-coagulants help to reduce this risk. These drugs can cause birth defects in developing babies. You must not become pregnant or make someone pregnant while taking these drugs.

If the myeloma comes back ([relapsed myeloma](#)) other [targeted therapies](#) may be used alone or in combination.

Chemotherapy for myeloma

[Chemotherapy](#) uses anti-cancer (cytotoxic) drugs to destroy cancer cells.

Chemotherapy drugs used to treat myeloma include:

- [melphalan](#)
- [cyclophosphamide](#)
- [bendamustine](#).

Steroids for myeloma

[Steroids](#) are drugs that can help destroy myeloma cells.

The 2 most commonly used steroids for myeloma are:

- prednisolone

- dexamethasone.

Steroids may be used on their own or in combination with other drugs. You usually take them as tablets. If you have difficulty swallowing them, you can have steroids that are liquid or that dissolve.

Steroids are also used to treat a possible [bone problem of myeloma](#), called spinal cord compression. This is when myeloma is in the bones of the spine and putting pressure on the spinal cord.

Having cancer drug treatment

Many cancer drugs for myeloma are taken as tablets or capsules by mouth (orally). Some may be given as an injection under the skin (subcutaneously). Others may be given by injection or drip (infusion) into a vein.

If you are having a drug by injection or drip (infusion) you usually have this at a day unit or clinic as an outpatient. For some treatments you might need to stay in hospital. You can take tablets or capsules at home.

Contraception and preventing pregnancy

It is important to follow your team's advice about contraception while you are having cancer drug treatment. This may include advice about:

- contraception to prevent pregnancy – if the drug might be harmful to the unborn baby
- barrier contraception to protect partners – if the drug might get into your body fluids.

If your treatment includes thalidomide, lenalidomide and pomalidomide, your team will give you extra information about contraception.

This is because these drugs can cause birth defects in developing babies. You must not become pregnant or make someone pregnant while taking these drugs.

Find out more in our information about:

- [thalidomide](#)
- [lenalidomide](#)
- [pomalidomide](#).

Intensive myeloma treatment

Sometimes, high dose chemotherapy with a [stem cell transplant](#) is used to treat myeloma. This is called intensive treatment. Stem cells are blood cells that are at the earliest stage of development. They are transplanted to help you recover from the side effects of treatment.

Intensive treatment may improve the chances of controlling the myeloma for a longer time.

If your doctor thinks this treatment is suitable for you, they will discuss it with you in more detail. Having this treatment will also depend on your general health. Stem cell transplants are only done in some specialist hospital units. You usually stay in hospital for a few weeks.

When this type of treatment is used to treat myeloma, it is more common to use your own stem cells for the transplant rather than stem cells from another person (a donor). Using your own stem cells is called an autologous stem cell transplant.

These are the different stages of an autologous stem cell transplant:

1. Induction treatment – you have a few cycles of chemotherapy and targeted therapy treatments to try to get rid of as many myeloma cells as possible.
2. Collecting stem cells – after the induction treatment, your stem cells are collected (harvested) and stored. This is done before you have high dose treatment. They are usually collected from the blood, but they can also be collected from the bone marrow.
3. High dose chemotherapy treatment – this aims to destroy any remaining myeloma cells and get the best response possible. But chemotherapy also affects healthy stem cells in your bone marrow and causes side effects.
4. Stem cell transplant – after the high dose treatment, the collected stem cells are given back to you through a drip (infusion). Your stem cells travel to your bone marrow and start making new healthy blood cells. This helps you to recover from the side effects of high dose treatment.

[Anthony Nolan](#) are a UK charity who provides information and support for people who need a stem cell transplant. They have a lot of detailed information you may find helpful.

Side effects of cancer drugs

Different drugs cause different side effects. Your cancer team will explain the possible side effects of any cancer drugs you are given.

You may get some of the side effects they mention, but you are very unlikely to get all of them. Although side effects can be hard to deal with, they usually disappear gradually when your treatment finishes.

It is important to tell your cancer team if you have any side effects while you are having treatment. They will give you 24-hour contact numbers for the hospital. If you feel unwell or need advice, you can call any time, day or night. Save these numbers in your phone or keep them somewhere safe.

Your cancer team may adjust your treatment to control side effects or to help to prevent more serious problems.

If you need information about a specific drug or combination, you can check our [cancer treatments and drugs A to Z](#). We have detailed information about many types of cancer drugs and their possible side effects.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Below is a sample of the sources used in our myeloma information. If you would like more information about the sources we use, please contact us at informationproductionteam@macmillan.org.uk

Hughes D, Yong K, Ramasamy K, et al. Diagnosis and management of smouldering myeloma: A British Society for Haematology Good Practice Paper. Br J Haematol. 2024;204(4):1193-1206. Available from: www.doi.org/10.1111/bjh.19333

Sive J, Cuthill K, Hunter H, et al. Guidelines on the diagnosis, investigation and initial treatment of myeloma: a British Society for Haematology/UK Myeloma Forum Guideline. Br J Haematol. 2021;193(2):245-268. Available from: www.doi.org/10.1111/bjh.17410

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Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

How we can help



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Get support from Macmillan Buddies

Sign up for our free Macmillan Buddies service and get 8 weeks of support. With a weekly call from one of our trained volunteers, you don't have to face cancer alone.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.

Get help